

## TASK 1

```
➤ mongo
Current Mongosh Log ID: 6a2e95f5813aed7b0dce5af
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.6.0
Using MongoDB:      4.0.12
Using Mongosh:       2.6.0

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

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The server generated these startup warnings when booting:
2026-09-19T09:14:02.074+05:30: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
2026-09-19T09:14:02.800+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
2026-09-19T09:14:02.800+05:30: This server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --bind_ip address to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning.
2026-09-19T09:14:02.800+05:30: Soft xlimits for open file descriptors too low
2026-09-19T09:14:02.800+05:30: For customers running the current memory allocator, we suggest changing the contents of the following sysfsfile
2026-09-19T09:14:02.800+05:30: We suggest setting the contents of sysfsfile to 0.
2026-09-19T09:14:02.800+05:30: Your system has glibc support for rseq built in, which is not yet supported by tcmalloc-google and has critical performance implications. Please set the environment variable GLIBC_TUNABLES=glibc.pthread.rseq=0

seq=0

test> use BookFlow
switched to db BookFlow
BookFlow> use BookFlow
already on db BookFlow
BookFlow> db.book_metadata.insertOne({
...   title: "The Silent Algorithm",
...   author: "A. Menon",
...   genre: "Technology",
...   publishedYear: 2022,
...   formats: ["hardcover", "Digital", "Audiobook"],
...   specs: {
...     pages: 340,
...     fileSize_MB: 4.2,
...     dmi: true,
...     narrator: "K. Rao",
...     runtime_min: 610
...   },
...   avgRating: 4.5,
...   reviews: [
...     {
...       member_id: "M1001",
...       rating: 5,
...       comment: "Excellent deep dive into digital systems design.",
...       date: ISODate("2023-01-15")
...     },
...     {
...       member_id: "M1042",
...       rating: 4,
...       comment: "Good but the digital edition had formatting issues.",
...       date: ISODate("2023-02-20")
...     }
...   ]
... })
```

```

... member_id: "M1042",
... rating: 4,
... comment: "Good! but the Digital Edition had formatting issues!",
... date: ISODate("2023-02-20")
... }
... })
{
  acknowledged: true,
  insertedId: ObjectId('6aae0684815a5d7b0d8ce5b0')
}
BookFlow> db.book_metadata.insertMany([
... {
...   title: "Ocean of Code",
...   author: "A. Fernandes",
...   genre: "Technology",
...   publishedYear: 2021,
...   formats: ["Digital", "Paperback"],
...   specs: {
...     pages: 210,
...     fileSize_MB: 2.1,
...     drm: false
...   },
...   avgRating: 4.2,
...   reviews: [
...     {
...       member_id: "M2001",
...       rating: 4,
...       comment: "Solid Digital read, clean layout.",
...       date: ISODate("2022-05-10")
...     },
...     {
...       member_id: "M2002",
...       rating: 5,
...       comment: "Best programming book this year.",
...       date: ISODate("2022-06-01")
...     }
...   ]
... },
... {
...   title: "Whispers of Yesterday",
...   author: "S. Kapoor",
...   genre: "Fiction",
...   publishedYear: 2019,
...   formats: ["Hardcover", "Paperback"],
...   specs: {
...     pages: 410,
...     binding: "Sewn"
...   },
...   avgRating: 3.8,
...   reviews: [
...     {
...       member_id: "M3001",
...       rating: 3,
...       comment: "Slow start but strong ending.",

```



```

    '2': ObjectId('6aae0699815a6d7b0d8ce5b3')
  }
}
BookFlow> db.book_metadata.countDocuments()
4
BookFlow> db.book_metadata.find().pretty()
[
  {
    _id: ObjectId('6aae0684815a6d7b0d8ce5b0'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: ['Hardcover', 'Digital', 'Audiobook'],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b1'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: ['Digital', 'Paperback'],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  }
]

```

```

{
  member_id: 'M2002',
  rating: 5,
  comment: 'Best programming book this year.',
  date: ISODate('2022-06-01T00:00:00.000Z')
}
],
{
  SS: S-3 SOA
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b2'),
    title: 'Whispers of Yesterday',
    author: 'S. Kopper',
    genre: 'Fiction',
    publishedYear: 2019,
    formats: [ 'Hardcover', 'Paperback' ],
    specs: { pages: 310, binding: 'Sewn' },
    avgRating: 3.8,
    reviews: [
      {
        member_id: 'M3001',
        rating: 3,
        comment: 'Slow start but strong ending.',
        date: ISODate('2020-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M3002',
        rating: 4,
        comment: 'Beautifully written prose.',
        date: ISODate('2020-02-11T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b3'),
    title: 'Quantum Horizons',
    author: 'P. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 540, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well-narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
}

```

## TASK 2:

```

BookFlow> db.book_metadata.find(
...   { avgRating: { $gt: 4 } },
..BTS{ title: 1, avgRating: 1, _id: 0 }
... )
[ Visualize capabilities
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
BookFlow>

```

```

BookFlow> db.book_metadata.find(
...   { formats: { $in: ["Digital", "Audiobook"] } },
...   { title: 1, formats: 1, _id: 0 }
... )
{
  title: 'The Silent Algorithm',
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
},
{
  title: 'Ocean of Code',
  formats: [ 'Digital', 'Paperback' ]
},
{
  title: 'Quantum Horizons',
  formats: [ 'Digital', 'Audiobook' ]
}
]
BookFlow>

```

```

BookFlow> db.book_metadata.find({
...   avgRating: { $gt: 4 },
...   formats: { $in: ["Digital"] }
... })
{
  _id: ObjectId('6aae0684815a6d7b0d8ce5b0'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: ISODate('2023-01-15T00:00:00.000Z')
    },
    {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: ISODate('2023-02-20T00:00:00.000Z')
    }
  ],
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2
}

```

```

]
BookFlow> db.book_metadata.find(
...   { formats: { $regex: "Digital", $options: "i" } },
...   { title: 1, formats: 1 }
... )
[ Run Spring Boot In VS Code
  {
    _id: ObjectId('6aae0684815a6d7b0d8ce5b0'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b1'),
    title: 'Ocean of Code',
    formats: [ 'Digital', 'Paperback' ]
  },
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b3'),
    title: 'Quantum Horizons',
    formats: [ 'Digital', 'Audiobook' ]
  }
]
BookFlow>

```

```

BookFlow> db.book_metadata.find(
...   { "reviews.comment": { $regex: "Digital", $options: "i" } },
...   { title: 1, "reviews.$": 1 }
... )
[
  {
    _id: ObjectId('6aae0684815a6d7b0d8ce5b0'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b1'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b3'),
    title: 'Quantum Horizons',
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      }
    ]
  }
]

```

TASK 3:

```
BookFlow> db.book_metadata.aggregate([
...   { $match: { publishedYear: { $gt: 2020 } } },
...   { $unwind: "$reviews" },
...   {
...     $group: {
...       _id: "$title",
...       avgReviewRating: { $avg: "$reviews.rating" },
...       totalReviews: { $sum: 1 },
...       author: { $first: "$author" }
...     },
...   },
...   { $sort: { avgReviewRating: -1 } },
...   {
...     $project: {
...       _id: 0,
...       title: "$_id",
...       author: 1,
...       avgReviewRating: { $round: [ "$avgReviewRating", 2 ] },
...       totalReviews: 1
...     }
...   }
... ])
[
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  },
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  },
  {
    totalReviews: 2,
    author: 'A. Fernandes',
    title: 'Ocean of Code',
    avgReviewRating: 4.5
  }
]
BookFlow> |
```

BreachPoint CTF 2 Users Teams Scoreboard

Members

User Name

nikanishpanikaj5 Captain

TASK 4:

```
BookFlow> db.book_metadata.createIndex({ "reviews.comment": "text" })
reviews.comment_text
BookFlow>
```

New folder Exercise 6 of le... Cybersec Skill... AI Tools Direct...

```
BookFlow> db.book_metadata.find(
...   { $text: { $search: "Digital" } },
...   { title: 1, score: { $meta: "textScore" } }
... ).sort({ score: { $meta: "textScore" } })
[
  {
    _id: ObjectId('6aae0684815a6d7b0d8ce5b0'),
    title: 'The Silent Algorithm',
    score: 1.1833333333333333
  },
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b3'),
    title: 'Quantum Horizons',
    score: 0.6
  },
  {
    _id: ObjectId('6aae0699815a6d7b0d8ce5b1'),
    title: 'Ocean of Code',
    score: 0.6
  }
]
BookFlow> |
```

```
BookFlow> db.book_metadata.find(
...   { $text: { $search: "Digital" } }
... ).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: 'CF6F4CEE',
    planCacheShapeHash: 'CF6F4CEE',
    planCacheKey: '08852285',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT MATCH'
    }
  }
}
```

```
BookFlow> db.book_metadata.createIndex({ formats: 1 })
formats_1
BookFlow>
```

```
BookFlow> db.book_metadata.createIndex({ avgRating: -1 })
avgRating_-1
BookFlow>
```